Health Risk Analysis using Machine Learning and Environmental Data

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Abstract

Air pollution has become a serious environmental issue in many urban regions, especially in cities such as delhi where pollution level is already reached hazardous limits. This study analyses large scale air quality data to understand pollution patterns across different locations. The dataset includes major pollutants such as $PM_{2.5}$, PM_{10}, NO_2, and SO_2 along with environmental factors like temperature, humidity, solar radiation, and rainfall, along with geographical attributes such as city, state, year, latitude, longitude etc. Initially machine learning based classification approach is used to analyze and model the relation between pollutant concentrations and air quality risk levels. A Random forest classifier is trained on the processed dataset to identify pollution risk categories based on environment conditions.The model achieves an accuracy around 72%, indicating its ability to capture important patterns in air quality data.

To improve interpretation of the results, an interactive visualization dashboard is developed using streamlit, to explore and analyze pollution trends across cities and states and observe variations in pollutant levels through graphical representations. In future work, deep learning models such as gradient boosting, XGBoost, and LSTM will be explored to improve prediction accuracy and supports pollution forecasting.

Keywords: Air pollution, Air quality analysis, Machine Learning, Random Forest, Environmental Data, Health risk prediction.

1 Introduction

Air pollution has become a major environmental and public health challenge, especially in rapidly growing urban areas. Cities like Delhi frequently experience high levels of air pollutants such as PM2.5, PM10, and other harmful gases, which significantly affect human health and overall quality of life. Changes in weather conditions such as temperature, humidity, and wind speed also influence air quality and the level of health risk for the people.

With the advancement of artificial Intelligence and Machine Learning, it has become possible to analyze environmental data and predict health risks more accurately. Machine Learning models can identify patterns in air quality and weather data to estimate the level of health risk faced by individuals.

In this research, a Random Forest based ml model is used to predict health risk levels using environmental parameters such as Air Quality Index (AQI), temperature, humidity, wind speed, solar radiation, and rainfall. The study also includes a dashboard for visualizing pollution patterns

and health risk predictions. The objective of this is to make decisions regarding environment health risks in polluted urban areas.

2 Literature Review

Air pollution has become an important area of research due to its serious environmental and health impacts. Researchers have used machine learning and deep learning models to analyze air quality data and predict about pollution and its impacts on health etc. Various studies have explored different models and approaches to understand the relationship between pollutants and environmental conditions.

Several studies have explored the use of machine learning models for predicting the AQI. Gupta and Kumar (2023) used machine learning algorithms to analyze environmental data for AQI prediction and shows that such techniques can effectively covers pollution patterns. Similarly, Kumar et al. (2024) proposed a regression-based approach for predicting hazardous $PM_{2.5}$ concentrations using environmental data from delhi. Although their model gives useful results but that was limited to a particular location.

Other researchers also explored different machine learning and deep learning approaches for air pollution prediction. Singh and Suthar (2025) compared machine learning and deep learning models for PM2.5 prediction, while Zhang et al. (2020) uses deep learning models such as LSTM to analyze pollution patterns. However, deep learning models requires high computational resources and may lack interpretability.

Devarakonda et al. (2018) further explored the use of machine learning methods for real time air quality prediction, highlights the potential of predictive systems for environmental monitoring.

Therefore, there is need to explore multiple models that can analyze multiple environmental parameters for pollution analysis. Since air pollution has a direct impact on human health, analyzing pollution patterns can also help in identifying health risks associated with poor or polluted air quality. In this study, machine learning and deep learning models are used to analyze air quality data and understand pollution patterns using multiple environmental features. As an initial approach, a Random Forest model is implemented due to its ability to understand complex environmental datasets, also supporting the analysis of pollution in relation to health issues or impacts.

Papers	**Existing Methodology**	**Limitations / Research Gaps**
Paper 1	ML models for AQI prediction	Limited models, no deep learning
Paper 2	Regression based PM2.5 prediction	Location specific, no hybrid models
Paper 3	Random Forest, Linear Regression	No real-time prediction
Paper 4	LSTM based deep learning	High complexity, low explainability
Paper 5	SVM, Decision Tree	Limited feature selection, lack of healthcare risk and multi-parameter analysis with health risk categorization

Table 1: Summary of Existing Literature and Research Gaps

3 Dataset Description

The dataset used in this research consists of air quality and environmental parameters collected from CPCB air monitoring sources of india. These datasets provide information about different air pollutants and meteorological conditions that influence air quality and health also. This data includes several important pollution indicators such as $PM_{2.5}$, PM_{10}, NO_2, SO_2, and O_3 along with multiple environmental and geographical features.

These parameters play an important role in determining aqi and understanding pollution patterns in urban areas. Since air pollution have serious impact on human health, analyzing these features can help in identifying health related risks with bad air quality.

Before applying different models, the dataset is preprocessed to remove missing values and ensure data consistency. The processed data is first used to analyze historical pollution patterns and understand their impact on human health. Based on the analysis, ml and dl models are used to predict future pollution levels.

Category	**Parameters**
Location Information	City, State, Latitude, Longitude
Time Features	Datetime, Year, Month, Day, Hour, Day of Week, Season
Weather Features	Temp_2m_C, Humidity_Percent, Dew_Point_C
Wind Features	Wind_Speed_10m_Kmh, Wind_Dir_10m, Wind_Gusts_Kmh
Rainfall Features	Precipitation_mm, Rain_mm, Is_Raining
Atmospheric Pressure	Pressure_MSL_hPa, Surface_Pressure_hPa
Solar and Cloud Data	Solar_Radiation, Cloud_Cover, Sunshine_Seconds
Pollution Features	PM2_5_ugm3, PM10_ugm3, CO_ugm3, NO2_ugm3, SO2_ugm3, O3_ugm3
AQI Indicators	US_AQI, EU_AQI, AQI_Category
Environmental Factors	Dust_ugm3, AOD
Special Conditions	Temp_Inversion, Festival_Period, Crop_Burning_Season

Table 2: Dataset Features Used for AQI Prediction

https://drive.google.com/file/d/1uk0XHNkWyjD3yYgd2coZOnFladmpBvcX/view?usp=drive_link

4 Proposed Methodology

The proposed system analyses air quality data to understand pollution patterns and their impact on human health. The data is preprocessed by removing missing values and cleaning the data. Important features such as pollutants ($PM_{2.5}$, PM_{10}, CO, NO_2, SO_2, O_3), weather parameters (temperature, humidity, windspeed, rainfall), and time-based factors are selected for analysis.

A machine learning model, initially Random Forest, is applied to learn patterns between environmental conditions and AQI levels. The results are visualized through an interactive dashboard to perform historical pollution analysis. Based on the learned patterns, the system is further used to predict future air pollution levels and identify health risks. In future work, additional machine learning models and features can be explored to improve prediction accuracy and forecasting.

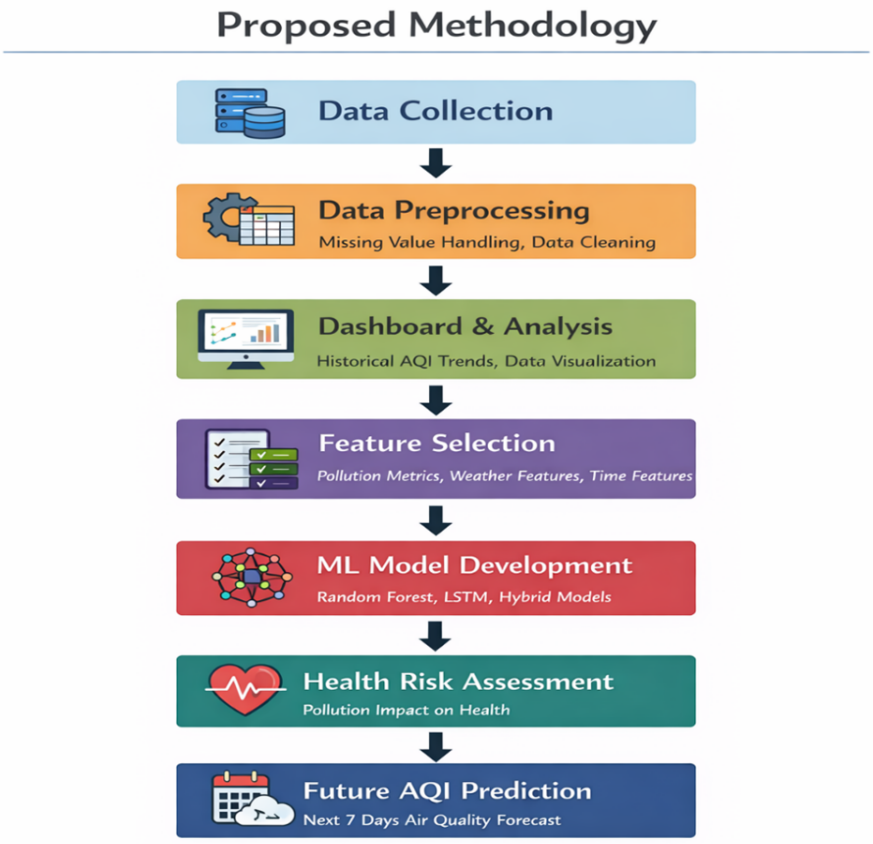


Figure 1: Proposed Methodology for AQI Prediction

5 Model Implementation

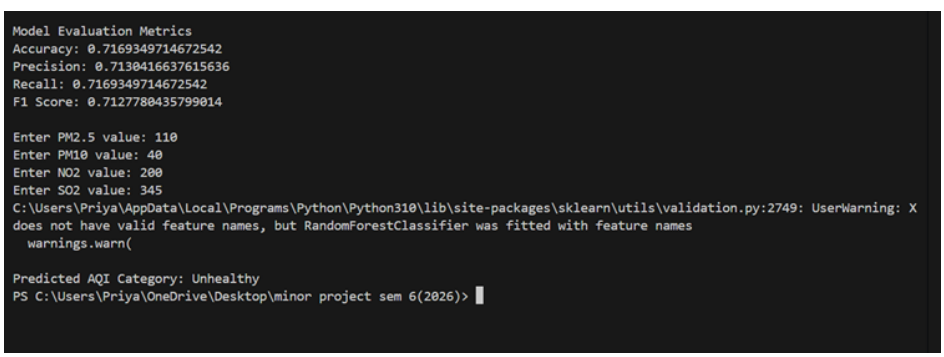
The Random Forest algorithm is used to model the relationship between environmental parameters and air quality. Random Forest is an ensemble learning method that builds multiple decision trees and combines their results to improve prediction accuracy and reduce overfitting.

The cleaned dataset, including pollutants, weather variables, and other environmental features, is used as input to train the model. The trained model learns patterns from historical data and helps predict AQI levels. These predictions can further be used to analyze pollution trends and support future air quality forecasting.

To evaluate the performance of the model, different evaluation metrics were used to measure how accurately the model predicts the results. Accuracy, precision, recall, and F1-score were considered to evaluate the effectiveness of the model. The dataset was divided into training and testing sets so that the model could be trained and evaluated on unseen data. This process ensures that the model performs consistently and avoids overfitting.

6 Results and Discussion

The results obtained from the Random Forest model provide useful insights into air pollution patterns. The model analyzes various environmental and pollution parameters to understand how different factors influence air quality levels. The results are visualized through an interactive dashboard that displays pollution trends, AQI categories, and environmental conditions using graphical representations.



```
Model Evaluation Metrics
Accuracy: 0.7169349714672542
Precision: 0.7130416637615636
Recall: 0.7169349714672542
F1 Score: 0.7127780435799014

Enter PM2.5 value: 110
Enter PM10 value: 40
Enter NO2 value: 200
Enter SO2 value: 345
C:\Users\Priya\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X
does not have valid feature names, but RandomForestClassifier was fitted with feature names
  warnings.warn(

Predicted AQI Category: Unhealthy
PS C:\Users\Priya\OneDrive\Desktop\minor project sem 6(2026)>
```

Figure 2: Results of the Random Forest Model

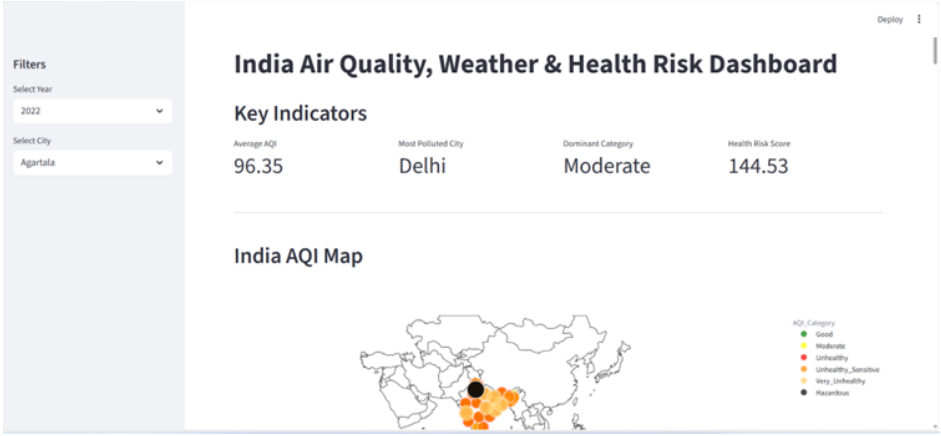


Figure 3: Interactive Dashboard for AQI Analysis

Figure 4: India AQI Map Visualization

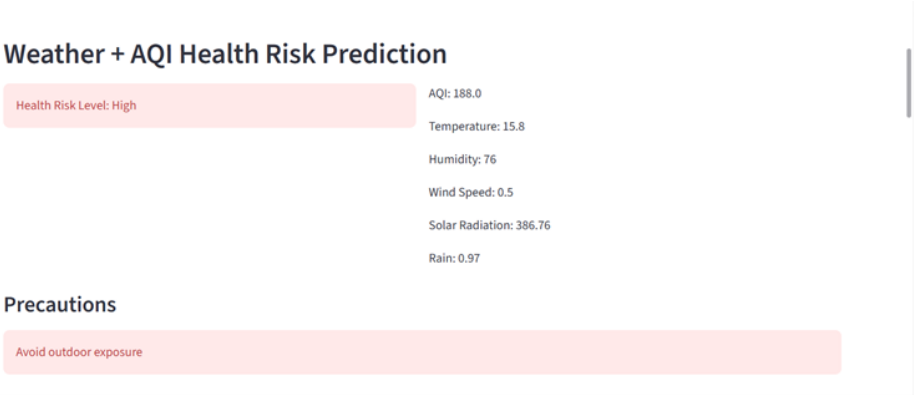


Figure 5: Health Risk Prediction based on AQI Levels

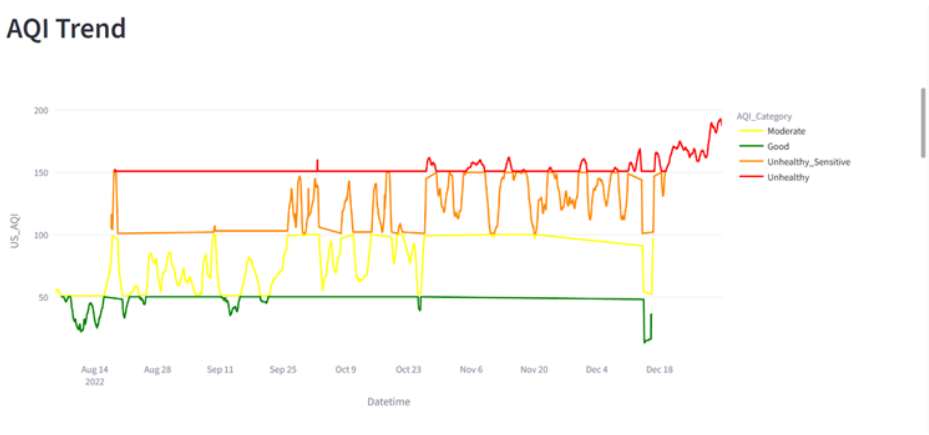


Figure 6: AQI Trend Analysis over Time

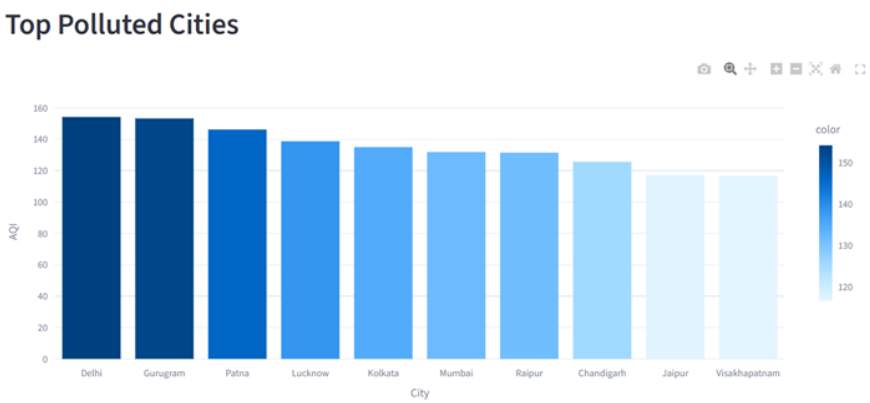


Figure 7: Top Polluted Cities based on AQI Levels

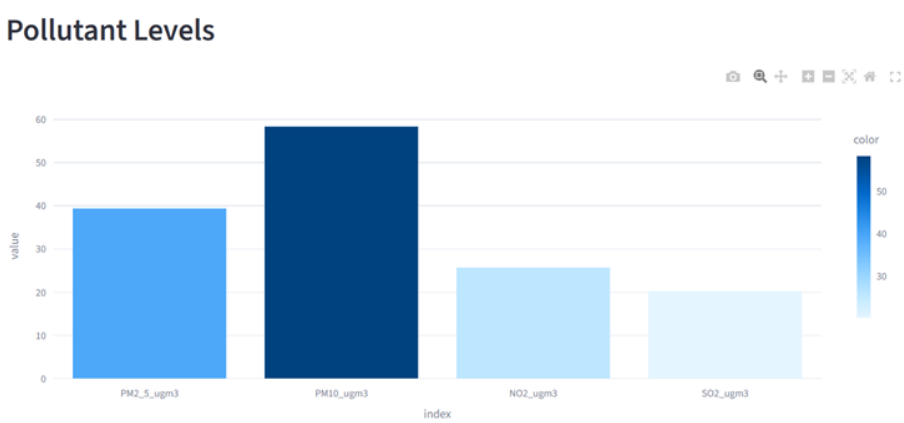


Figure 8: Pollutant Levels Analysis of Major Air Pollutants

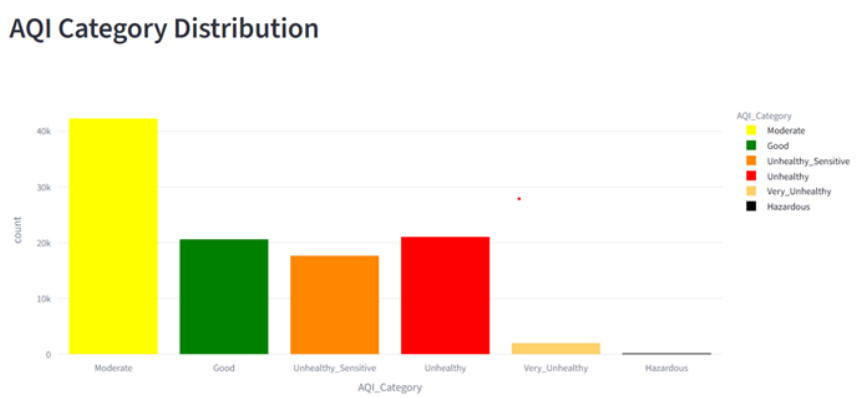


Figure 9: Distribution of AQI Categories

City Ranking Table

City	US_AQI
Delhi	154.1605
Gurugram	153.2333
Patna	146.191
Lucknow	138.6827
Kolkata	134.9448
Mumbai	131.8328
Raipur	131.4366
Chandigarh	125.6029
Jaipur	117.212
Visakhapatna	116.8507

Take care of your health and stay informed about air quality levels in your city!

Figure 10: City Ranking based on AQI Levels

Through the dashboard, users can explore historical pollution data and observe variations in AQI across different time periods and environmental conditions. These visualizations help in identifying pollution patterns and understanding environmental and health-related impacts. These findings can be used in future work to develop models for predicting upcoming air quality conditions.

7 Conclusion and Future Scope

This research study analyzes air pollution data to understand pollution patterns and classify health risk using machine learning techniques. A Random Forest model was used to train the dataset and achieved approximately 72% accuracy. An interactive dashboard was also developed using Streamlit to visualize environmental parameters and pollution trends. The results show that machine learning can effectively assist in analyzing environmental pollution and identifying potential health risks.

In future work, more advanced machine learning and deep learning models such as LSTM, XGBoost, and Gradient Boosting can be explored to improve prediction accuracy. The dashboard can also be enhanced to allow users to input environmental parameters and obtain predictions of future air quality and health risk levels for upcoming days. Additional visualization features and real-time data integration can further make the system more informative and practical.

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